Vehicle Architecture and Structural Design

Design objective

Stable, recoverable 25 in vehicle with a 4,000 ft simulated apogee target.

Structures Lead responsibilities

- Von Kármán nose and shoulder fit
- Fin-can and body integration
- Recovery attachment and load path
- PLA build and fit checks

Vehicle-level tradeoffs

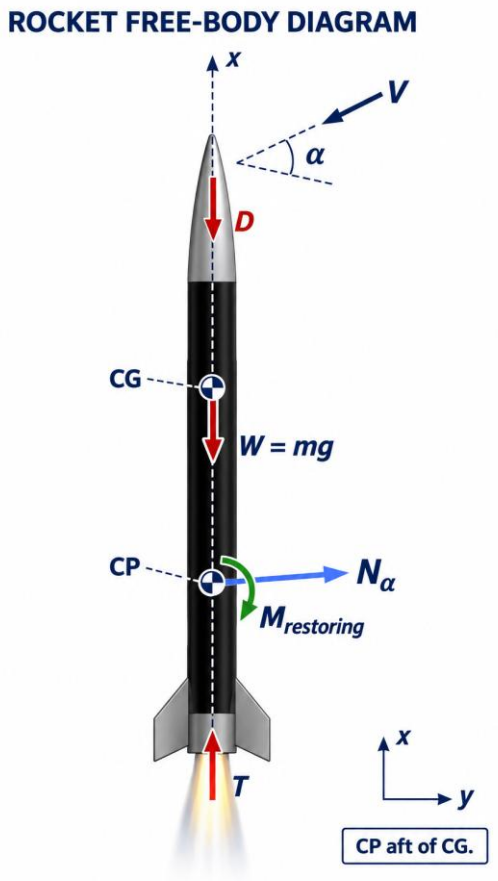
- Nose mass changes CG and stability
- Retention must allow motor ejection
- Avionics and recovery share tube space

CAD assembly



Nominal geometry; black body tube

Forces and static stability



Modeled CG ahead of CP; force directions shown

OpenRocket: 4,086 ft simulated apogee | 1.23-caliber modeled static margin | Built and ground-tested; never flown

Nose Cone, Shoulder Fit, and Separation

Figure 1. Dimensioned Von Kármán nose and body joint

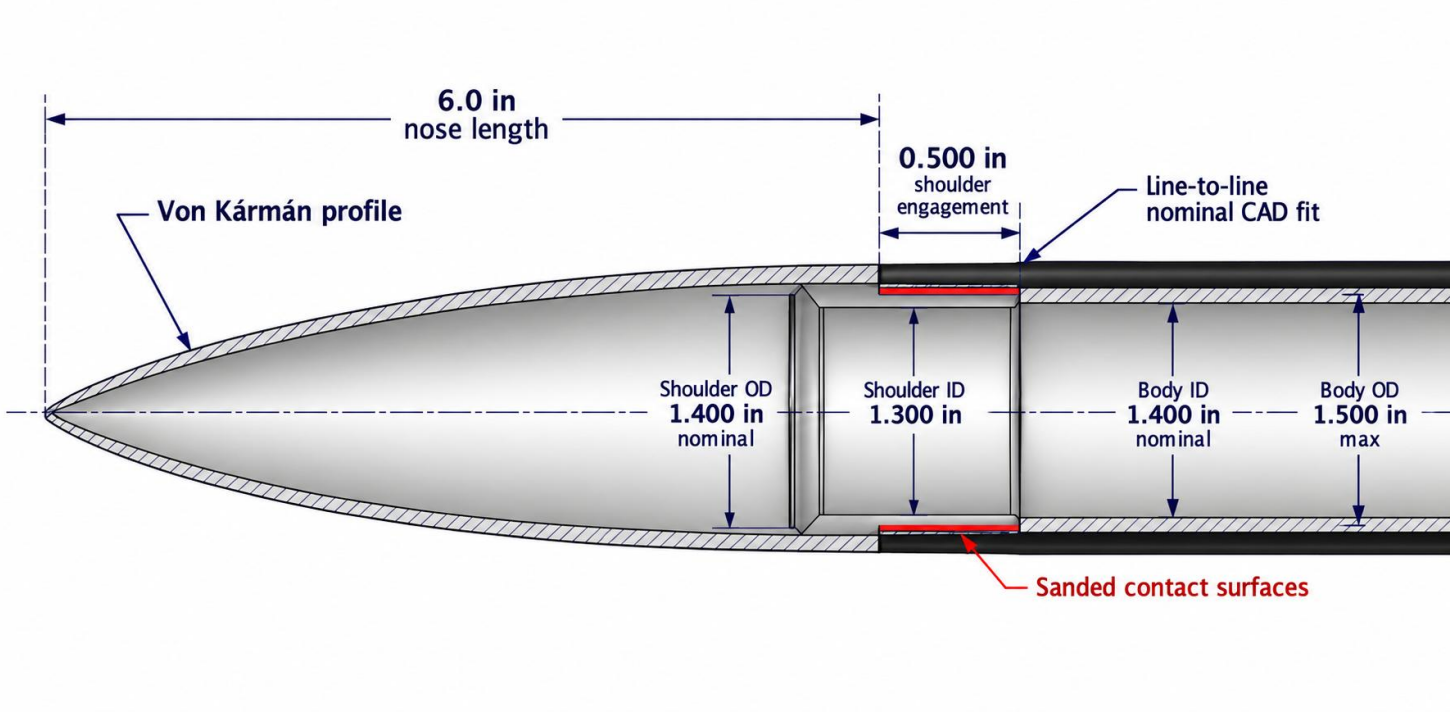
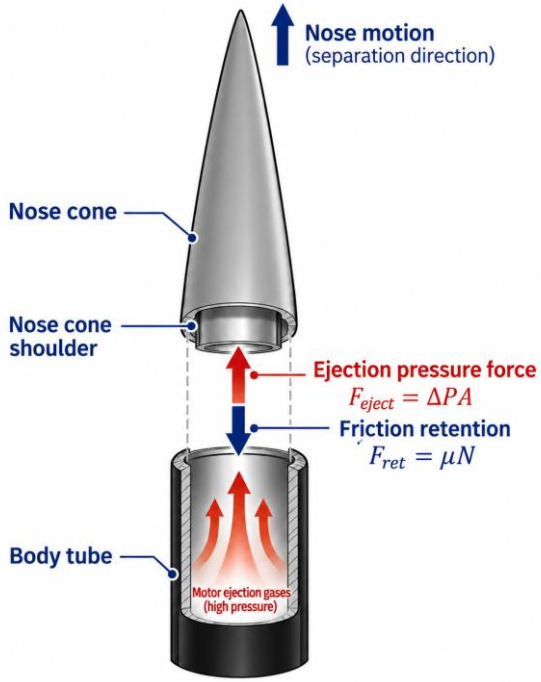


Figure 2. Idealized separation forces



Fit and pressure checks

Nominal clearance: 1.400 − 1.400 = 0.000 in; radial wall: 0.050 in.

Recalled extraction force ≈19 N after sanding the printed shoulder.

Ideal nose pressure: 19 N / 9.931 × 10^{-4} m^2 = 19.1 kPa (2.77 psi).

System constraint

Pressure uses the exposed nose area. A sealed avionics bay needs a verified gas route to the recovery cavity; bypass losses require an ejection test.

Recovery Eyelet: Geometry and Structural Comparison

Original-style eyelet

Wider rounded profile

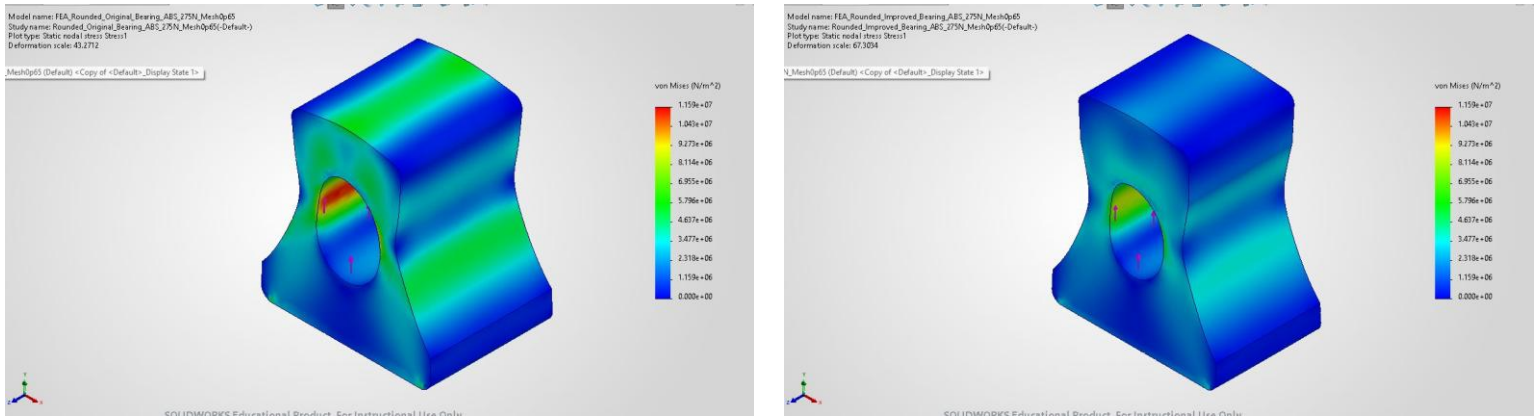


Figure 3. Common 0–11.591 MPa stress scale; saved SolidWorks results at 0.65 mm mesh target.

Applied load and fixed-root reaction

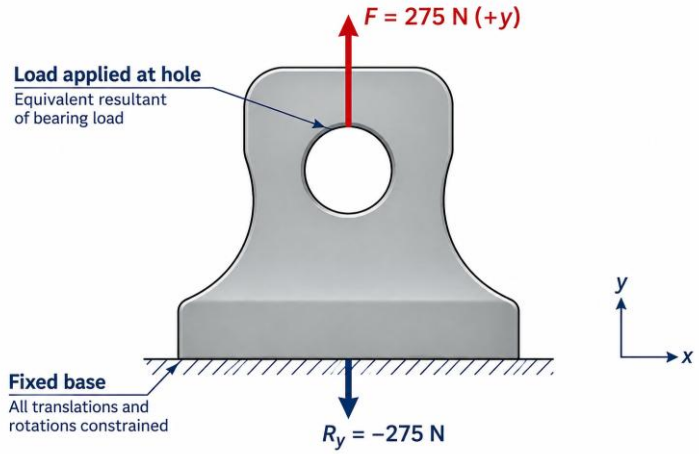
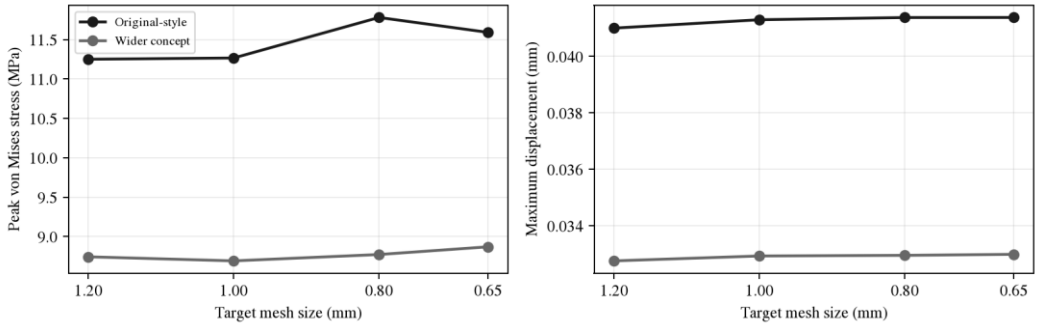


Figure 4. Equivalent 275 N hole load and root restraint.

Mesh refinement, both geometries



Four 275 N mesh sizes. From 0.80 to 0.65 mm: peak stress changed about 1.6%; displacement <0.2%.

Key findings and design lessons

Peak stress: 11.591 to 8.869 MPa (23.5% lower)

Displacement: 0.04138 to 0.03299 mm (20.3% lower)

- Wider, smoother profile reduced modeled peak stress.
- Recovery, structure and packaging change together.
- Printed PLA, pin contact and ejection need physical tests.

FEA comparison uses an idealized fixed root and assumed load.